Capstone hypothesis and baseline

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Why must a checkpoint chosen using validation data still face an untouched final test?

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Write a testable capstone hypothesis and a credible baseline. Try the recall before opening notes.

Your learning target

I can write a testable capstone **hypothesis** and a credible **baseline**.

I can specify data roles and a selection rule before inspecting results.

I can calculate **baseline** MSE and distinguish a **success criterion** from a result.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Documenting a network experiment. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Write a testable capstone **hypothesis** and a credible **baseline**

Specify data roles and a selection rule before inspecting results

Calculate **baseline** MSE and distinguish a **success criterion** from a result

Retrieve what you already know

> Why must a checkpoint chosen using validation data still face an untouched final test?

My starting explanation

This week's end goal

Explain capstone **hypothesis** and **baseline** with a traceable toy-network case and a bounded conclusion.

Notice the evidence

> Investigate the supplied Capstone hypothesis and baseline case. Separate observations, given facts and assumptions before choosing an explanation.

A capstone needs a question that **evidence** can answer. “Build a better network” leaves better undefined.

Here the question is: does two-update full-batch **fitting** of one scalar ReLU weight reduce validation mean squared error relative to the initial-weight **baseline** on a constructed positive-**input** task?

Predict max(0, a*x), with no bias. The **baseline** has a = 1. The trained candidate starts at a = 1, uses learning rate 0.1 and exactly two updates. These choices are written before results are inspected.

The **hypothesis** is that the trained candidate has lower validation MSE. The plan does not guarantee that result. This task tests **training** and record keeping, not whether neural networks are preferable to every other model.

A stronger comparison may also include an ordinary fitted line.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Capstone hypothesis and baseline. Connect each labelled input, deciding step and result.

Role and ID	x	y
Training T1	1	2
Training T2	2	4
Validation V1	3	6
Validation V2	4	8
Final test	Sealed	Sealed

Training rows determine updates. Validation rows compare the prespecified candidate with the **baseline**. The final-test **input**s and targets remain sealed until the procedure and reporting plan are frozen.

MSE = sum of squared **prediction** errors divided by the number of evaluated rows. **Baseline prediction**s on V1 and V2 are 3 and 4. Errors are -3 and -4, squared errors 9 and 16, and validation MSE is 25/2 = 12.5.

There are two validation cases.

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
T1 and T2	Fit two updates	Training result only
V1 and V2	Compare with a = 1	Validation comparison
Sealed final rows	Evaluate frozen procedure	Final evidence on those rows

Each arrow preserves a distinct data role. Opening final rows to choose a learning rate changes their role.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Capstone hypothesis and baseline. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Compute **baseline prediction**s, errors, squared errors and MSE on V1 and V2. State the denominator.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

> **Change one thing**
>
> Change: Give only B the eight evaluation labels during **training**.

> **Predicted result**
>
> **Prediction**: The proposed comparison becomes contaminated rather than a valid held-out test.

Why the result follows

Reason: One version has gained privileged information from the reserved evaluation set.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Capstone hypothesis and baseline to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Hypothesis	A testable expectation with a stated comparison and outcome
Baseline	A specified reference procedure evaluated on the same cases
Success criterion	A rule written before evaluation for judging the measured result
Reserved test	Cases kept out of fitting, tuning and selection

Use two terms to explain a step in this week’s example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Capstone hypothesis and baseline, predict a step, then compare with its explanation.

Worked explanation · Capstone hypothesis and baseline

The validation criterion is candidate MSE strictly below **baseline** MSE on V1 and V2. Equality does not pass. The final criterion is lower MSE than the same **baseline** on the sealed final rows.

Report both scores and every raw **prediction** whether or not the criterion passes. Record the rule, **training** rows, update count, learning rate, loss, data versions and code version.

A criterion specifies how to judge a future result; it is not **evidence** that the result already happened.

Keep validation and final criteria separate. Do not silently add epochs or change pre**process**ing after seeing a weak result. A changed plan is a new version and needs an honest account of what **evidence** informed it.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone hypothesis and baseline

A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

The facts we will use

Supplied facts

Two frozen versions are planned: A uses corner features only; B also uses the centre feature.

Eight reserved toy patterns have labels defined before scoring.

AI ACADEMY / AI-5/33

Capstone hypothesis and baseline

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Write the prediction

B should have more correct labels than A on the same eight patterns.

2. Declare the baseline

A is a documented corner-only network, not a deliberately broken version.

3. Reserve evaluation

Keep the eight patterns out of model selection and tuning.

CHANGE ONE THING

Give only B the eight evaluation labels during training.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Write the prediction	B should have more correct labels than A on the same eight patterns.
2. Declare the baseline	A is a documented corner-only network, not a deliberately broken version.
3. Reserve evaluation	Keep the eight patterns out of model selection and tuning.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A capstone **hypothesis** links a specific architectural change to a measured outcome. A **baseline** anchors the comparison; frozen labels and held-out patterns protect it from answer-key tuning.

What this example does not establish

Keep the stated limit

Eight toy patterns support a narrow experiment, not claims about real photographs, children or universal architecture superiority.

Keep the stated limit

Read the labelled picture: Name the intended change, the **baseline** and the measured outcome. Why must both versions avoid **training** on the eight reserved labels?

Keep the stated limit

Change one thing: Give only B the eight evaluation labels during **training**. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Capstone hypothesis and baseline claim and explain a repair using the supplied evidence.

Check the claim against the facts

Supported example: a separate **baseline** predicts 2 and 4 for validation targets 5 and 7. Its errors are -3 and -3 and MSE is 9.

A proposed plan says to train one fixed candidate and report both MSE values on those same two rows. It supplies no candidate **prediction**s, so no improvement can yet be claimed.

Guided: compute that baseline MSE, name the missing evidence, and state whether the hypothesis has been confirmed.

Locate the earliest unsupported step

A proposal says “try models until the test score looks good”. Identify the missing fixed choices and explain why that test is not independent.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Capstone hypothesis and baseline with support, then explain the deciding step in your own words.

State the **prediction** rule, **baseline**, candidate learning rate and update count, comparison data and metric. Write a testable **hypothesis**.

Guided: compute that baseline MSE, name the missing evidence, and state whether the hypothesis has been confirmed.

Explain your prediction to a partner before comparing results.

Practise with less help: Calculate the reference score

Use workbook W2 for the connected example “Capstone **hypothesis** and **baseline**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Capstone **hypothesis** and **baseline**. Proposed explanation: “A **hypothesis** is a result”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Name the intended change, the **baseline** and the measured outcome. Why must both versions avoid **training** on the eight reserved labels?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Capstone hypothesis and baseline. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Compute **baseline prediction**s, errors, squared errors and MSE on V1 and V2. State the denominator.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Capstone hypothesis and baseline. Explain why each step is justified.

Explain → apply → challenge

State the **prediction** rule, **baseline**, candidate learning rate and update count, comparison data and metric. Write a testable **hypothesis**.

A proposal says “try models until the test score looks good”. Identify the missing fixed choices and explain why that test is not independent.

A partner inspects final labels and changes the learning rate. Explain how to document this and obtain independent **evidence**.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: A new task predicts max(0, a*x), with **baseline** a = 1.

Training rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed. Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates.

Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

Keep actual **observation**s separate from expected results, preserve failures, compare a **baseline** and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Before running, exchange a plan with a partner. The partner should be able to identify what is changed, what stays fixed, what metric is computed, which rows select the procedure and when the final test may be opened.

A file name alone does not protect a split; record who can inspect it and how related examples were separated.

> **Exit: someone opens the final targets to choose between learning rates, then calls the winning score independent final evidence. Explain the change of role and the next honest step.**

A partner inspects final labels and changes the learning rate. Explain how to document this and obtain independent **evidence**.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Capstone hypothesis and baseline case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
T1 and T2	Fit two updates	Training result only
V1 and V2	Compare with a = 1	Validation comparison
Sealed final rows	Evaluate frozen procedure	Final evidence on those rows

Each arrow preserves a distinct data role. Opening final rows to choose a learning rate changes their role.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Plan a new case independently

A new task predicts max(0, a*x), with **baseline** a = 1. **Training** rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed.

Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A new task predicts max(0, a*x), with **baseline** a = 1. **Training** rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed.

Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

> **My independent procedure, result and limitation**
>
> | | |
> |---|---|
> | | |
> | | |
> | | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Capstone hypothesis and baseline, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

By the end, I can	How I will show it
Write a testable capstone hypothesis and a credible baseline	Explain the deciding step, show a trace or test, and state one limit.
Specify data roles and a selection rule before inspecting results	Explain the deciding step, show a trace or test, and state one limit.
Calculate baseline MSE and distinguish a success criterion from a result	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Capstone hypothesis and baseline without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Write a testable capstone **hypothesis** and a credible **baseline**

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Keep the eight patterns out of model selection and tuning.

1. Reserve evaluation

2. Declare the **baseline**

3. Write the **prediction**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reserve evaluation Keep the eight patterns out of model selection and tuning. A capstone **hypothesis** links a specific architectural change to a measured outcome.

A **baseline** anchors the comparison; frozen labels and held-out patterns protect it from answer-key tuning.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Give only B the eight evaluation labels during **training**. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The proposed comparison becomes contaminated rather than a valid held-out test.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The proposed comparison becomes contaminated rather than a valid held-out test. One version has gained privileged information from the reserved evaluation set.

Repair a missing building block

Review Documenting a network experiment (ai-5/32). Goal: Record a network experiment with enough detail to replay its calculations. Short practice: What additional **evidence** is needed for a final performance claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dataset shift in neural models (ai-5/30). Goal: Explain how a changed **input** mix can change overall accuracy. Short practice: Did the supplied model's dim-light accuracy deteriorate?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Similarity and normalisation (ai-5/27). Goal: Calculate vector length and normalise a nonzero vector. Short practice: Why do the original two metrics select different winners?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

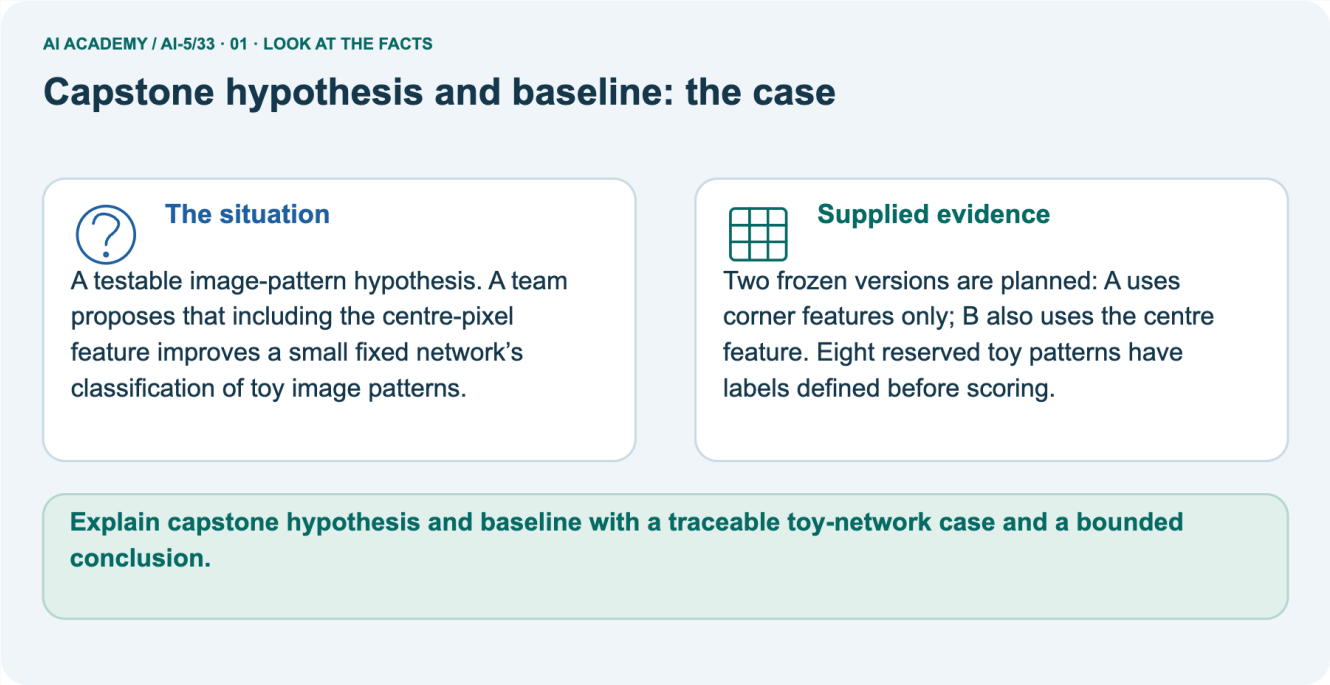
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Capstone hypothesis and baseline graphic using its labels, deciding evidence and relevant limitation.

01. Capstone hypothesis and baseline: the case



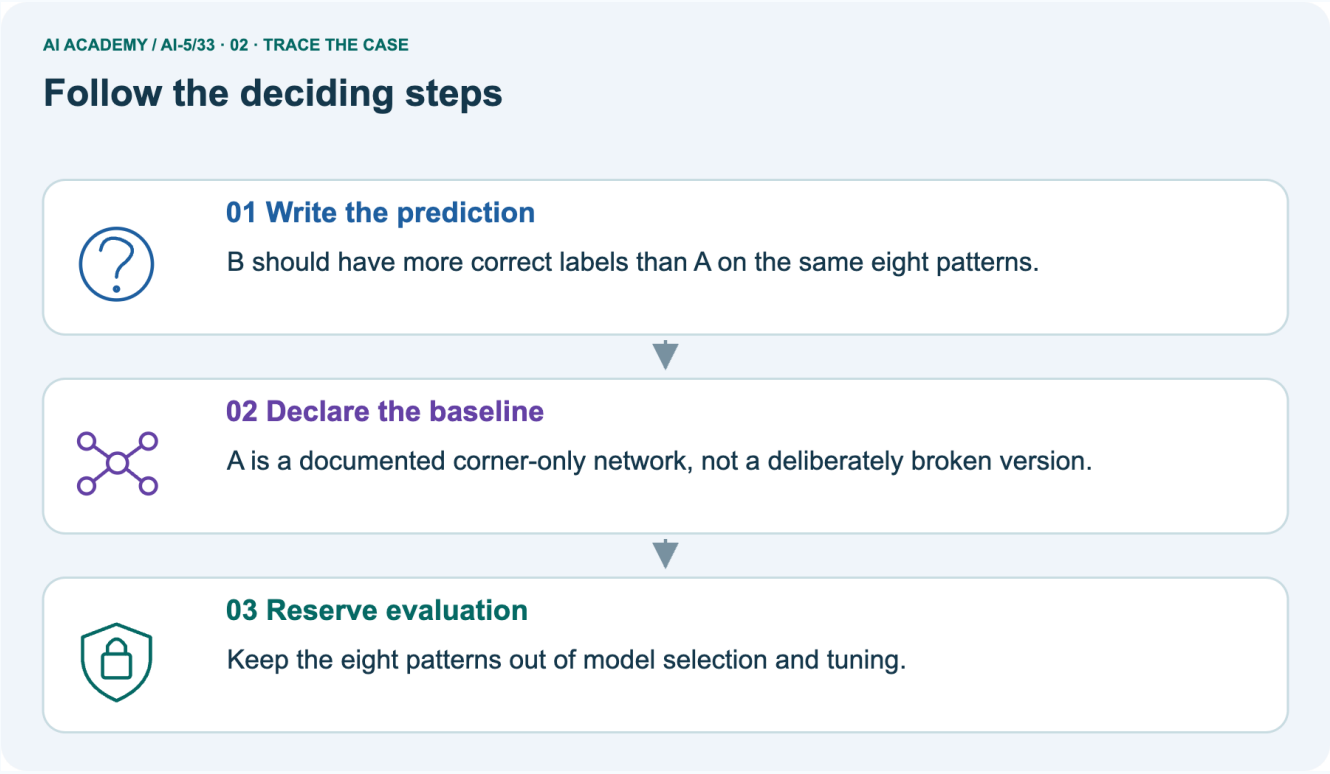
Follow the labelled picture with the matching explanation. Open the vector diagram

A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

Supplied facts: Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

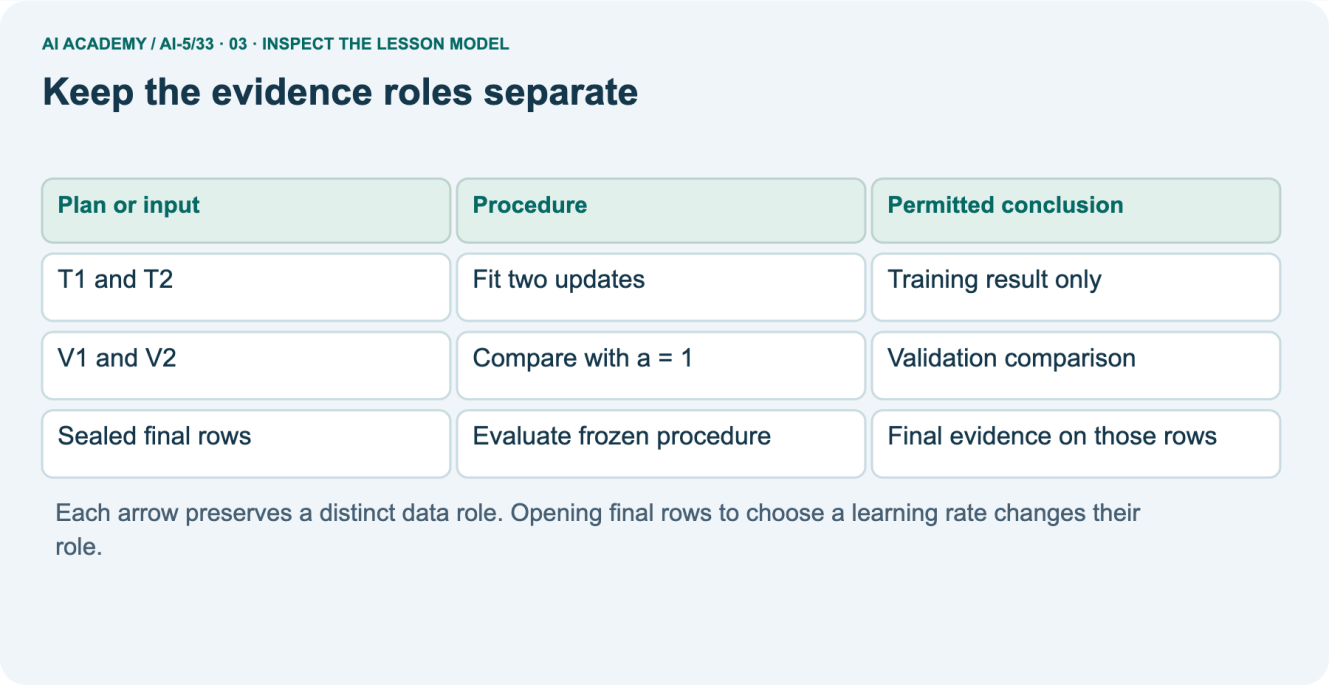


Follow the labelled picture with the matching explanation. Open the vector diagram

Write the **prediction**: B should have more correct labels than A on the same eight patterns. → Declare the **baseline**: A is a documented corner-only network, not a deliberately broken version. → Reserve evaluation: Keep the eight patterns out of model selection and tuning.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Keep the evidence roles separate

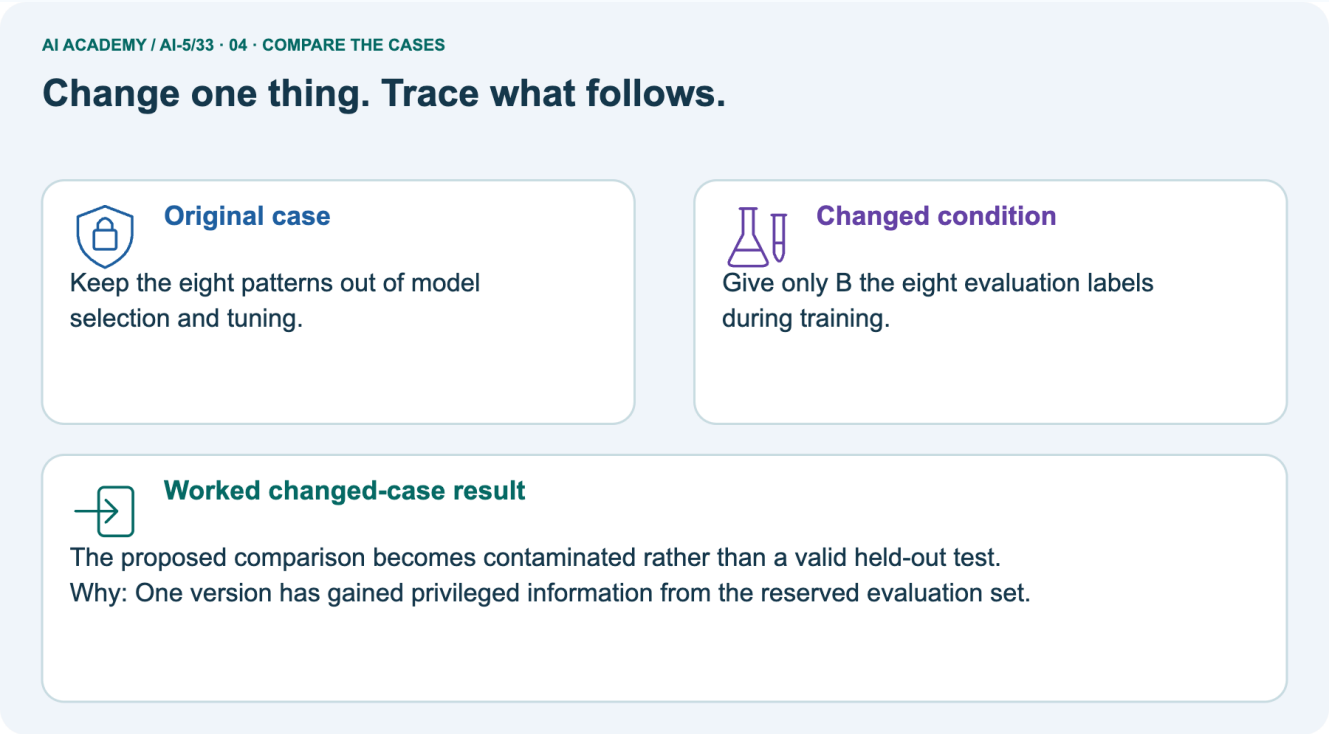


Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the **evidence** roles separate. T1 and T2 | Fit two updates | **Training** result only. V1 and V2 | Compare with a = 1 | Validation comparison. Sealed final rows | Evaluate frozen procedure | Final **evidence** on those rows

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



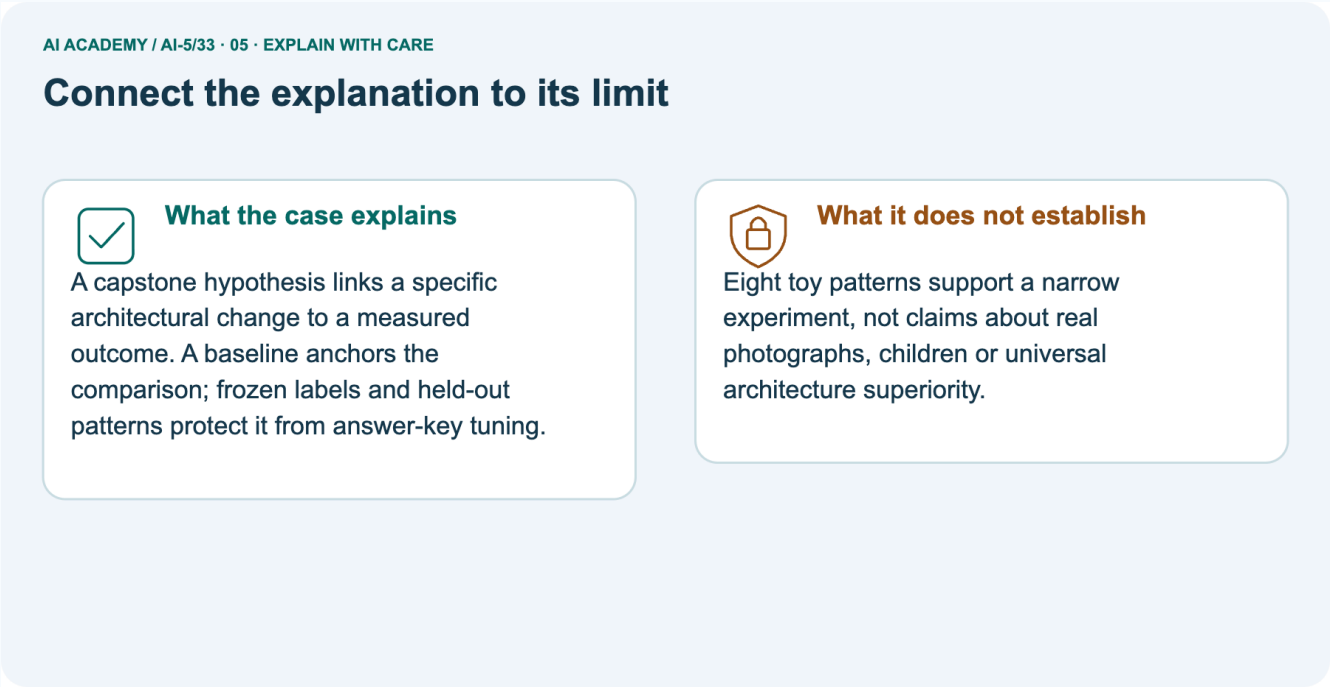
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Keep the eight patterns out of model selection and tuning. Changed condition: Give only B the eight evaluation labels during **training**.

Expected result: The proposed comparison becomes contaminated rather than a valid held-out test. Reason: One version has gained privileged information from the reserved evaluation set.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



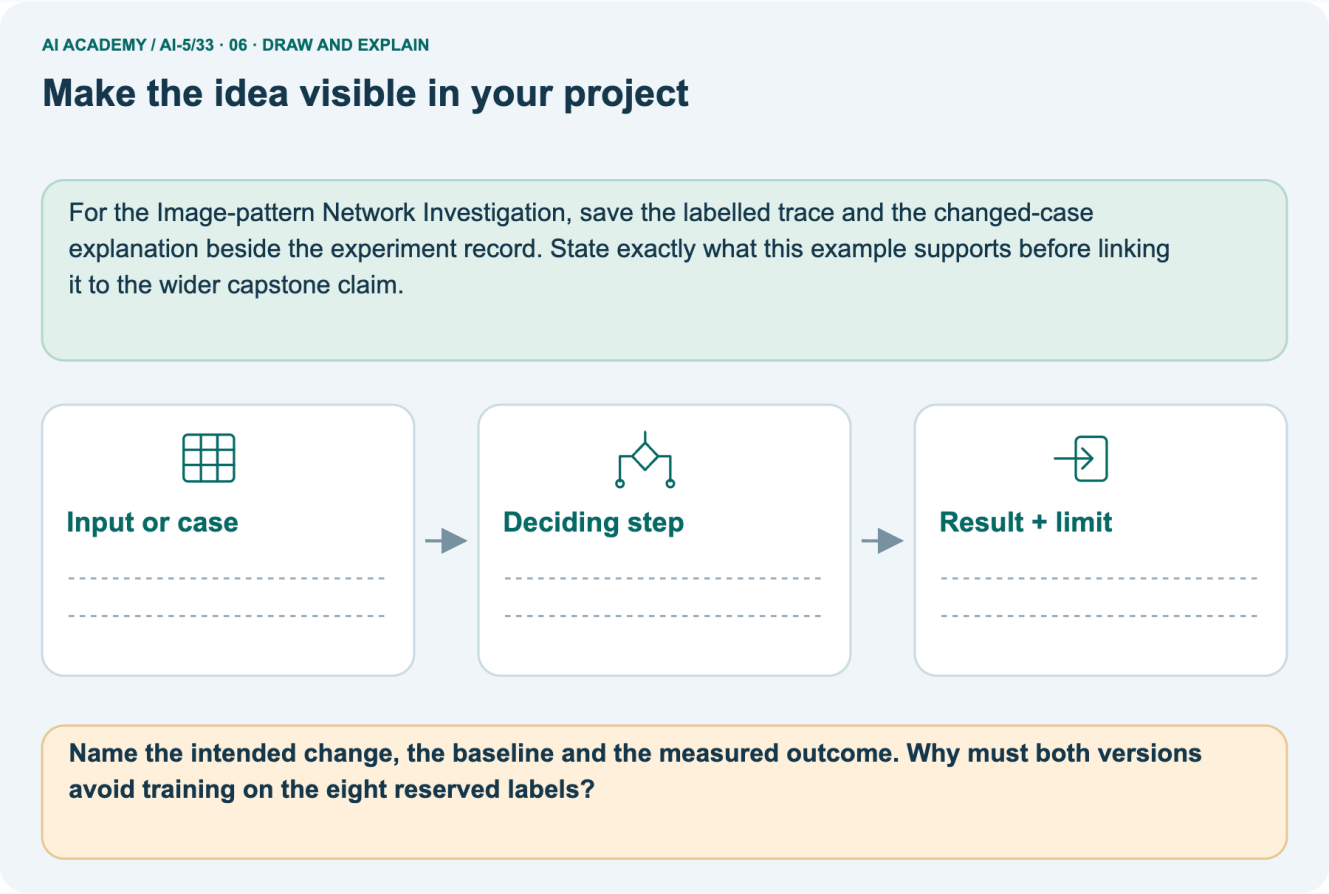
Follow the labelled picture with the matching explanation. Open the vector diagram

A capstone **hypothesis** links a specific architectural change to a measured outcome. A **baseline** anchors the comparison; frozen labels and held-out patterns protect it from answer-key tuning.

Limit: Eight toy patterns support a narrow experiment, not claims about real photographs, children or universal architecture superiority.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Name the intended change, the **baseline** and the measured outcome. Why must both versions avoid **training** on the eight reserved labels?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.